DSA Course Kickoff: Understanding Binary Numbers

Course Overview

- **New Course** : **Data Structures and Algorithms (DSA)** .
- **Duration** : The course will run for the next **6 months** .
- **Level** : Covers topics from **Basic to Advanced** .
- **Instructor** : **Rohit Negi** .
- **Instructor's Background** :
- Completed **B.Tech** from a **Tier 3 college** .
- Cleared **GATE CSE** with a good rank.

Understanding the Binary Number System

- **Purpose of Binary** :
- Binary numbers (**0** and **1**) directly represent the states of **transistors** inside a computer.
- **On** state is represented by **1** .
- **Off** state is represented by **0** .

- **Limitations of the Old Decimal System (with Bulbs)** :
- Previously, each decimal number from **0-9** was represented by an **individual bulb** .
- For example, to show '4', the **'4' bulb** would turn on.
- **Major Limitation** : This system could only represent **0-9 unique numbers** at a time.
- **Problem with Multi-digit Numbers** : It couldn't clearly represent numbers like '73' because it would be confusing (could it be '37'?).

- **Scientists' Solution: Embracing Binary** :
- Scientists decided to remove decimal numbers and use only **bulbs (transistors)** to represent information.
- **Example: Representing the number '13'** :

1. First, **13** is converted into its binary form: **1101** .
2. This binary number is then represented using bulbs: The **rightmost bulb** is **On** , the next is **Off** , then **On** , then **On** . Any remaining bulbs are **Off** .
3. When the computer reads this sequence (e.g., **0000001101**), it converts it back to **13** in decimal.

- **Key Advantage** : Binary allows computers to represent **many more numbers** in the same physical space compared to the old system of 10 unique decimal bulbs. The use of **transistors (0/1)** fundamentally changed computing.

Transistor Capacity and Data Storage

- **Representing Conditions** :
- Just **two bulbs (transistors)** can represent **4 unique conditions** : **00, 01, 10, 11** .
- These conditions correspond to decimal numbers **0, 1, 2, 3** .
- **Computer's Language** : Computers inherently read and process information in **binary (0s and 1s)** .

Key Takeaways

- The upcoming **DSA course** will provide comprehensive learning from **basic to advanced** over **6 months** .
- The **binary number system** is crucial for computers because it directly maps to the **On/Off states of transistors** .
- Binary efficiently solved the problem of representing **multi-digit numbers** that the older decimal-bulb systems couldn't handle.
- Even a small number of transistors can represent a **wide range of values** , making computing powerful and efficient.